Dingtek Sensor Integration Guide

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Service portal: https://cloud.dingtek.com

1. Purpose and scope

This guide explains the supported ways to integrate Dingtek wireless sensors with either a customer-managed platform or the Dingtek Cloud service.

It covers:

- Direct reporting from cellular or Wi-Fi sensors to a customer server;
- LoRaWAN integration through a LoRaWAN network server;
- Access to devices, attributes, and telemetry through the Dingtek Cloud HTTPS API;
- API access limits, data retention, availability, and commercial terms.

Product-specific payload formats, command definitions, credentials, and sample decoders are supplied separately when applicable. Contact Dingtek Sales or your Dingtek project representative for the files applicable to your order.

2. Integration options

2.1 Customer-managed platform

Cellular and Wi-Fi products, including NB-IoT, LTE Cat 1, LTE Cat M, 4G, and Wi-Fi models, may be configured to report directly to a customer-managed server when supported by the product firmware.



The customer must provide the required server information, protocol selection, credentials, and network requirements before configuration. Dingtek may assist with initial endpoint configuration before shipment when this service is included in the order.

For this integration model:

- The customer operates and secures the receiving server;
- The customer is responsible for availability, storage, backups, and application development;
- Dingtek provides the applicable device protocol and command documentation;
- Sample code or decoders may be provided as reference material, but the customer remains responsible for testing and production implementation.

2.2 LoRaWAN integration

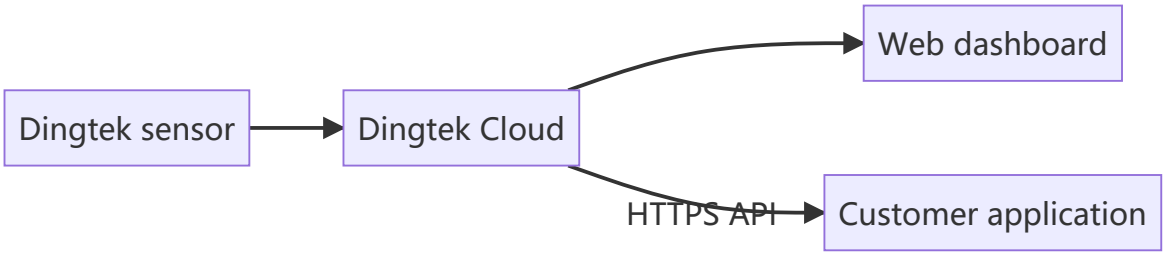
LoRaWAN sensors first communicate with a compatible LoRaWAN gateway. The gateway forwards traffic to a LoRaWAN network server such as ChirpStack, The Things Stack, or another compatible platform.



Gateway coverage, radio planning, regional frequency settings, network-server configuration, and device commissioning are project-specific responsibilities unless explicitly included in the Dingtek quotation or contract.

2.3 Dingtek Cloud

Dingtek Cloud can be used for evaluation, pilot projects, or production service under a paid service plan. It provides device management, dashboards, charts, reports, attributes, and telemetry access.



All customer access to Dingtek Cloud must use the HTTPS domain:

```
https://cloud.dingtek.com
```

Do not integrate with a server IP address or a direct backend port. IP addresses and internal ports may change or may be blocked without notice. Plain HTTP requests may be redirected, but integrations must be implemented with HTTPS directly.

3. Sensor reporting to a customer server

Protocol documents are provided after an applicable order is placed. Contact Dingtek Sales to request:

- The protocol document for the exact product and firmware version;
- Uplink and downlink payload definitions;
- Command examples;
- Available Python, JavaScript, ChirpStack, or other sample code;
- Supported endpoint configuration options.

The customer must validate the integration with test devices before production deployment. Protocol support and configurable connection parameters vary by model and firmware version.

4. LoRaWAN devices connected to Dingtek ChirpStack

Where MQTT access is included and enabled for the project, an MQTT client can subscribe to device events or publish downlink commands through the Dingtek-managed ChirpStack service.

Typical ChirpStack topic formats are:

```
application/APPLICATION_ID/device/DEV_EUI/event/up
```

```
application/APPLICATION_ID/device/DEV_EUI/command/down
```

Replace `APPLICATION_ID` and `DEV_EUI` with the values assigned to the project. MQTT hostname, TLS requirements, port, username, password, topic permissions, and payload format are provided separately. Do not assume that anonymous MQTT access is available.

MQTT is normally more suitable than periodic API polling when low-latency event delivery is required. LoRaWAN devices assigned to Dingtek Cloud may also expose stored telemetry through the HTTPS API, subject to the service plan and retention policy.

5. Dingtek Cloud HTTPS API

5.1 API documentation and base URL

Swagger UI is available at:

https://cloud.dingtek.com/swagger-ui/

The API base URL is:

```
https://cloud.dingtek.com/api
```

Common endpoints include:

```
POST /api/auth/login
GET  /api/user/devices
GET  /api/plugins/telemetry/{entityType}/{entityId}/values/attributes
GET  /api/plugins/telemetry/{entityType}/{entityId}/values/timeseries
```

Available endpoints and permissions depend on the account role and the installed Dingtek Cloud version.

5.2 Authentication

Use the account supplied by Dingtek. Never place a real username, password, or bearer token in source control, public documentation, screenshots, or shared logs.

Linux/macOS example:

```
curl --request POST \
  --url 'https://cloud.dingtek.com/api/auth/login' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data '{"username":"YOUR_ACCOUNT","password":"YOUR_PASSWORD"}'
```

Windows PowerShell example:

```powershell
$body = @{
    username = "YOUR_ACCOUNT"
    password = "YOUR_PASSWORD"
} | ConvertTo-Json -Compress

$response = Invoke-RestMethod `
    -Method Post `
    -Uri "https://cloud.dingtek.com/api/auth/login" `
    -ContentType "application/json" `
    -Headers @{ Accept = "application/json" } `
    -Body $body
```

A successful login returns authentication tokens. Use the returned access token as a bearer token for subsequent requests. Tokens must be stored securely and refreshed or replaced when they expire.

5.3 Querying the assigned device list

```curl
curl --request GET \
  --url 'https://cloud.dingtek.com/api/user/devices?
pageSize=100&page=0&type=DEVICE_MODEL&sortProperty=name' \
  --header 'Accept: application/json' \
  --header 'X-Authorization: Bearer YOUR_ACCESS_TOKEN'
```

Replace:

- `DEVICE_MODEL` with the applicable device type, such as `DF703`, when a type filter is required;
- `YOUR_ACCESS_TOKEN` with the access token returned by the login request.

Use pagination when more devices exist than the requested `pageSize`. Do not assume that one response contains every device.

5.4 Querying device attributes

```curl
curl --request GET \
  --url
'https://cloud.dingtek.com/api/plugins/telemetry/DEVICE/YOUR_DEVICE_ID/values/at
tributes' \
  --header 'Accept: application/json' \
  --header 'X-Authorization: Bearer YOUR_ACCESS_TOKEN'
```

Replace `YOUR_DEVICE_ID` with the device UUID returned by the device-list API. A device name, IMEI, DevEUI, serial number, and device UUID are not interchangeable.

5.5 Querying historical telemetry

```
curl --get \
  --url
'https://cloud.dingtek.com/api/plugins/telemetry/DEVICE/YOUR_DEVICE_ID/values/timeseries' \
  --header 'Accept: application/json' \
  --header 'X-Authorization: Bearer YOUR_ACCESS_TOKEN' \
  --data-urlencode 'keys=temperature,humidity' \
  --data-urlencode 'startTs=START_TIME_IN_MILLISECONDS' \
  --data-urlencode 'endTs=END_TIME_IN_MILLISECONDS' \
  --data-urlencode 'limit=1000'
```

`startTs` and `endTs` are Unix timestamps in milliseconds. Request only the keys and time range needed, paginate or divide large time ranges into smaller requests, and verify the response before advancing to the next range.

6. API access and rate limits

6.1 Login limit

The login endpoint is limited by source IP address:

```
5 login requests per minute per source IP
```

Requests above the permitted rate may return:

```
HTTP 429 Too Many Requests
```

Applications must not repeatedly submit credentials. Cache and securely reuse valid tokens until renewal is required. Implement exponential backoff after authentication failures or HTTP 429 responses. Multiple users behind the same NAT, office gateway, or VPN may share one public source IP and therefore share the same limit.

6.2 General API fair use

Clients must use reasonable polling intervals, request only necessary fields and time ranges, and avoid concurrent retry loops. Dingtek may temporarily throttle or block traffic that threatens platform stability, appears abusive, causes excessive load, or violates the agreed service scope.

Dingtek reserves the right to rate-limit, block, suspend, or terminate access when abnormal activity is detected. Abnormal activity includes, but is not limited to, excessive request frequency, uncontrolled automated retries, credential attacks, attempts to bypass security controls, malicious requests, unauthorized use, or activity that may affect the security or availability of the platform or other customers. These controls may be applied to an IP address, credential, user account, customer account, device, integration, or other access channel as appropriate.

Unless a separate written agreement states otherwise, the service does not guarantee a fixed request volume, throughput, latency, or concurrency level for general API endpoints. Customers requiring high-volume or low-latency delivery should discuss MQTT, dedicated integration, or a separate service plan with Dingtek Sales.

7. Data retention and customer backups

Telemetry and related historical data are retained for **up to one month**. Data older than one month may be deleted automatically and may not be recoverable.

The one-month period is a maximum retention target, not a guarantee that every record will remain available for the full period. Data may also be lost or removed because of maintenance, device re-registration, account or device deletion, service faults, storage management, or other operational reasons.

Customers who require long-term records must export data regularly to their own storage. Dingtek Cloud must not be treated as the customer's only archive or backup system unless a separate written backup and retention agreement has been signed.

Deleting a customer, user, or device may also delete or detach associated data. Contact Dingtek before deleting production entities if historical data must be preserved.

8. Availability, maintenance, and incident handling

Dingtek operates the shared cloud service on a best-effort basis. If the server is unavailable or an outage occurs, Dingtek will use commercially reasonable efforts to investigate and restore service.

Unless a separate written service-level agreement has been signed:

- No uninterrupted availability or uptime percentage is guaranteed;
- No specific incident response or repair time is guaranteed;
- Maintenance may be performed when required for security, reliability, capacity, or upgrades;
- API behavior, security controls, or platform functions may change when necessary;
- Dingtek does not provide service credits, compensation, or reimbursement for downtime, delayed recovery, unavailable data, or resulting business loss.

Customers are responsible for designing their systems to tolerate temporary network or service interruption, including local buffering, controlled retry with backoff, duplicate handling, timestamp preservation, and reconciliation after recovery.

9. Commercial terms and support scope

Dingtek Cloud is a **paid service and is not provided free of charge**. Pricing, account duration, device quantity, enabled functions, traffic expectations, support scope, and renewal terms depend on the applicable quotation or contract. Contact Dingtek Sales for current commercial terms.

Unless explicitly included in writing, the cloud service does not include custom software development, customer-side debugging, third-party platform maintenance, guaranteed data recovery, unlimited API traffic, or a dedicated service instance.

Customers should complete renewal no later than **one week before the service expiration date** to avoid interruption. If the service remains unpaid or unrenewed for more than one week after its expiration date, Dingtek reserves the right to stop the service and delete the associated customer accounts, user accounts, devices, configurations, telemetry, historical records, and other related data. Deleted data may not be recoverable.

The customer is responsible for tracking the service term, completing payment and renewal on time, and exporting any required data before expiration. Dingtek is not obligated to retain data for an expired or unrenewed service unless a separate written agreement states otherwise.

Accounts or services may also be suspended or terminated because of non-payment, misuse, security risk, abnormal or malicious access, or violation of agreed terms.

10. Customer responsibilities and security

The customer must:

- Keep account passwords, API tokens, MQTT credentials, and device credentials confidential;
- Store all credentials and cryptographic keys securely and prevent their disclosure, sharing, or unauthorized use;
- Change any credential that has been exposed in email, chat, screenshots, code, or logs;
- Use HTTPS and supported secure protocols;
- Restrict access according to least-privilege principles;
- Validate timestamps, units, payload versions, missing values, and duplicate records;
- Monitor integration errors and avoid uncontrolled retry loops;
- Notify Dingtek promptly of suspected unauthorized access;
- Maintain its own exports, backups, and business-continuity procedures.

The customer is responsible for all access performed with its accounts, passwords, tokens, API keys, MQTT credentials, device keys, certificates, and other credentials. To the extent permitted by applicable law, Dingtek is not responsible for unauthorized access, data disclosure, service misuse, device operation, financial loss, business loss, or other damage resulting from the customer's failure to secure, rotate, restrict, or promptly report compromised credentials.

Dingtek may revoke or rotate credentials, restrict an account or source IP, apply rate limits, block requests, suspend service, or terminate access when reasonably necessary to protect customers or the platform, enforce the agreed service scope, or respond to abnormal or malicious activity.

11. HTTP response guidance

Common responses include:

Status	Meaning	Recommended action
`200`	Request succeeded	Validate and process the response
`400`	Invalid request or JSON	Check syntax, parameters, and timestamp units
`401`	Missing, expired, or invalid authentication	Check credentials or renew the token without rapid retries
`403`	Account lacks permission	Confirm account role and device assignment
`404`	Endpoint or entity not found	Check the URL, entity type, and device UUID
`429`	Rate limit exceeded	Stop retries and apply exponential backoff
`5xx`	Temporary server-side failure	Retry later with bounded exponential backoff

12. Change history

Version	Date	Summary
2.0	20 August 2026	Updated all cloud access to HTTPS/domain-based URLs; corrected API and command examples; replaced broken local images with portable diagrams; documented login rate limiting, abnormal-access controls, credential responsibility, retention, renewal and post-expiration deletion, availability, commercial terms, security, and customer responsibilities
1.x	Earlier release	Initial integration guidance